5. Implement a program based on using Tuples, Lists, Dictionaries, Sets.

```
### Student Enrollment Program

# List to store students (each student is a tuple)
students = []

# Set to track unique courses
courses = set()

# Dictionary to track enrolled courses for each student
enrollment = {}


# Function to add a student
def add_student(name):
  student = (name,)        # Create a tuple for the student
  students.append(student)   # Add the tuple to the list
  enrollment[name] = []     # Initialize enrolled courses
  print(f"Added student: {name}")


# Function to enroll a student in a course
def enroll_student(name, course):
  if name in enrollment:
    enrollment[name].append(course)
    courses.add(course)    # Add course to the set
    print(f"{name} enrolled in {course}.")
  else:
    print(f"Student {name} not found.")


# Function to display all students and their courses
def display_students():
  if not students:
    print("No students enrolled.")
  else:
    print("\nStudents and their courses:")

    for student in students:
      name = student[0]
      enrolled_courses = enrollment[name]

      if enrolled_courses:
        print(f"{name}: {', '.join(enrolled_courses)}")
      else:
        print(f"{name}: No courses enrolled")


# Main program
def main():
  while True:
    print("\nMenu:")
    print("1. Add Student")
    print("2. Enroll Student in Course")
    print("3. Display Students")
    print("4. Exit")

    choice = input("Enter your choice (1-4): ")
```

```
        if choice == '1':
            name = input("Enter student name: ")
            add_student(name)

        elif choice == '2':
            name = input("Enter student name: ")
            course = input("Enter course name: ")
            enroll_student(name, course)

        elif choice == '3':
            display_students()

        elif choice == '4':
            print("Exiting the program.")
            break

        else:
            print("Invalid choice. Please try again.")


# Start the program
if __name__ == "__main__":
    main()
```

Output:

```
Menu:
1. Add Student
2. Enroll Student in Course
3. Display Students
4. Exit
Enter your choice (1-4): 1
Enter student name: John
Added student: John

Menu:
1. Add Student
2. Enroll Student in Course
3. Display Students
4. Exit
Enter your choice (1-4): 2
Enter student name: John
Enter course name: mca
 John enrolled in mca
Menu:
1. Add Student
2. Enroll Student in Course
3. Display Students
4. Exit
Enter your choice (1-4): 3
Students and their courses:
John: mca
```

6. Implement a program based on using Functions, Function with parameters.

```
def add_numbers(a, b):
    """Calculate and return the sum of two numbers."""
    return a + b

def multiply_numbers(a, b):
    """Calculate and return the product of two numbers."""
    return a * b

def main():

    num1 = float(input("Enter the first number: "))
    num2 = float(input("Enter the second number: "))

    sum_result = add_numbers(num1, num2)
    product_result = multiply_numbers(num1, num2)

    print(f"The sum of {num1} and {num2} is: {sum_result}")
    print(f"The product of {num1} and {num2} is: {product_result}")

if __name__ == "__main__":
        main()
```

Output:

```
Enter the first number:  10
Enter the second number:  20
The sum of 10.0 and 20.0 is: 30.0
The product of 10.0 and 20.0 is: 200.0
```

7. Implement a program based on Functions Inside of Functions.

```
# Basic Arithmetic Operations Program

def arithmetic_operations():

  def add(a, b):
    return a + b

  def multiply(a, b):
    return a * b

  while True:
    print("\nChoose an operation:")
    print("1. Addition")
    print("2. Multiplication")
    print("3. Exit")

    choice = input("Enter your choice (1-3): ")

    if choice == '1':
      num1 = float(input("Enter first number: "))
      num2 = float(input("Enter second number: "))

      result = add(num1, num2)
      print(f"The result of addition is: {result}")

    elif choice == '2':
      num1 = float(input("Enter first number: "))
      num2 = float(input("Enter second number: "))

      result = multiply(num1, num2)
      print(f"The result of multiplication is: {result}")

    elif choice == '3':
      print("Exiting the program.")
      break

    else:
      print("Invalid choice. Please try again.")
```

```
# Run the main function
if __name__ == "__main__":
    arithmetic_operations()
```

Output:

```
Choose an operation:
1. Addition
2. Multiplication
3. Exit
Enter your choice (1-3): 1
Enter first number: 10
Enter second number: 5
The result of addition is: 15.0

Choose an operation:
1. Addition
2. Multiplication
3. Exit
Enter your choice (1-3): 2
Enter first number: 10
Enter second number: 5
The result of multiplication is: 50.0

Choose an operation:
1. Addition
2. Multiplication
3. Exit
Enter your choice (1-3): 3
Exiting the program.
```

8. Implement a program based on built-in functions .

```
def main():
    numbers = input("Enter numbers separated by spaces: ")
    number_list = list(map(int, numbers.split()))
    total_sum = sum(number_list)
    max_value = max(number_list)
    min_value = min(number_list)
    sorted_list = sorted(number_list)

    # Output the results
    print(f"Sum: {total_sum}")
    print(f"Maximum: {max_value}")
    print(f"Minimum: {min_value}")
    print(f"Sorted List: {sorted_list}")
if __name__ == "__main__":
    main()
```

Output:

```
Enter numbers separated by spaces:  10 20 30
Sum: 60
Maximum: 30
Minimum: 10
Sorted List: [10, 20, 30]
```